

D597

Data Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A customer table at a 60-year-old hardware distributor stores phone numbers as '555-0101, 555-0199' in a single Phone column for customers with two numbers. The table has 48,000 rows and is backed up nightly. Which normal form does this design violate?

- A. First normal form (1NF)
- B. Third normal form (3NF)
- C. Second normal form (2NF)
- D. None, because there is still one row per customer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An Orders table has key OrderID and includes CustomerID and CustomerCity. CustomerCity depends on CustomerID, not on OrderID. Which normal form is violated, and what is the fix?

- A. 2NF; split the composite key.
- B. No violation, because every column relates to the order somehow.
- C. 1NF; split the city name into atomic parts.
- D. 3NF; move CustomerCity to a Customers table keyed by CustomerID.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A retail analytics team's nightly sales dashboard, built by a contractor last spring, joins seven normalized tables and takes 14 minutes to load. The underlying data changes only once a day. The team's warehouse runs on hardware purchased in 2021. Which change best speeds reads while accepting a known trade-off?

- A. Never denormalize, because redundancy is always a design error.
- B. Normalize further to remove the remaining redundancy.
- C. Denormalize into a wide reporting table (for example a star schema), accepting redundancy refreshed daily.
- D. Replace the 2021 hardware.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A clinic database has Patients(PatientID primary key, Name) and Visits(VisitID primary key, PatientID, VisitDate). The clinic sees about 90 patients a day and uses a cloud-hosted server. A clerk tries to insert a visit with PatientID 7731, which does not exist in Patients. Visits.PatientID is a foreign key. What happens?

- A. The insert succeeds with PatientID set to NULL.
- B. The insert succeeds, because foreign keys are checked only when a parent is deleted.
- C. The insert succeeds and a new patient 7731 is created automatically.
- D. The insert is rejected because it would break referential integrity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Customers has 500 rows. Orders has 1,200 rows placed by 420 distinct customers, all of whom exist in Customers. The marketing team plans to email everyone after the holiday sale, which runs 10 days. How many rows does Customers LEFT JOIN Orders ON CustomerID return?

- A. 1,700
- B. 1,280
- C. 1,200
- D. 500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An INNER JOIN of Products (300 rows) to Sales on ProductID returns 2,000 rows. A LEFT JOIN from Products to Sales returns 2,045 rows. How many products have never sold?

- A. 0, because an inner join already includes every product
- B. 255
- C. 45
- D. 2,045

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst at a 3,000-seat call center wants departments whose average handle time exceeds 8 minutes, counting only calls from 2025. The query is scheduled to run at 6 a.m. Which query structure is correct?

- A. GROUP BY dept ORDER BY AVG(handle_min) DESC LIMIT 8
- B. GROUP BY dept HAVING call_year = 2025 AND AVG(handle_min) > 8
- C. WHERE AVG(handle_min) > 8 AND call_year = 2025 ... GROUP BY dept
- D. WHERE call_year = 2025 ... GROUP BY dept HAVING AVG(handle_min) > 8

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A Tickets table has 10 rows. The Resolver column is NULL in 3 of them. What do COUNT(*) and COUNT(Resolver) return?

- A. 7 and 10
- B. 10 and 7
- C. 10 and 10
- D. 10 and 3

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A data team at a regional credit union with 14 branches needs to add a new LoyaltyTier column to the existing Members table before the third-quarter campaign. The database administrator prefers work done on Tuesdays. Which kind of statement and command does this require?

- A. DDL: ALTER TABLE Members ADD LoyaltyTier ...
- B. DML: UPDATE Members SET LoyaltyTier ...
- C. DML: INSERT INTO Members ...
- D. DDL: CREATE TABLE LoyaltyTier ...

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10. [TRANSFER — reversed stem / different variable]

Which statement removes only the rows whose Status is 'closed' and keeps the table and its other rows?

- A.** TRUNCATE TABLE Tickets
- B.** ALTER TABLE Tickets DROP COLUMN Status
- C.** DELETE FROM Tickets WHERE Status = 'closed'
- D.** DROP TABLE Tickets

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A payroll system at a 2,500-employee manufacturer runs one transaction that debits the company account \$80,000 and credits employee accounts \$80,000. The system was upgraded in March. Power fails after the debit but before the transaction commits. After restart, what does atomicity guarantee?

- A.** The March upgrade logs let an admin finish the credits by hand.
- B.** The debit is rolled back, so neither the debit nor the credits take effect.
- C.** The credits are applied automatically on restart because of durability.
- D.** The debit stays and is flagged for review.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A media startup with 35 employees in two time zones stores user profiles whose fields vary widely: some users have 3 attributes, others 60. The app almost always reads a whole profile by user ID. The CTO previously worked at a bank. Which storage approach fits best?

- A.** A document database storing each profile as one document keyed by user ID.
- B.** A graph database.
- C.** A relational database, because the CTO knows banking systems.
- D.** A relational table with a column for every possible attribute.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A customer table at a ~~60-year-old hardware distributor~~ stores phone numbers as '555-0101, 555-0199' in a single Phone column for customers with two numbers. ~~The table has 48,000 rows and is backed up nightly.~~ Which normal form does this design violate?

Target word: which normal form **Decides:** One column holds a list of values, so values are not atomic.

A	Correct	1NF requires atomic values; a list of phone numbers in one field breaks it.
B	Confusion set	3NF is about transitive dependencies between non-key columns.
C	Adjacent category	2NF is about partial dependency on a composite key, which is not the issue.
D	Dropped element	One row per customer does not fix the non-atomic column.

2. Answer D Concept 1 · transfer

An Orders table has key OrderID and includes CustomerID and CustomerCity. CustomerCity depends on CustomerID, not on OrderID. Which normal form is violated, and what is the fix?

Target word: name the violation from the dependency **Decides:** A non-key column depends on another non-key column (transitive).

A	Confusion set	The key is a single column, so there is no partial dependency.
B	Dropped element	Ignores that the city depends on the customer, not the order.
C	Adjacent category	The city value is already atomic.
D	Correct	OrderID → CustomerID → CustomerCity is a transitive dependency.

3. Answer C Concept 2

A retail analytics team's nightly sales dashboard, ~~built by a contractor last spring,~~ joins seven normalized tables and takes 14 minutes to load. The underlying data changes only once a day. ~~The team's warehouse runs on hardware purchased in 2021.~~ Which change best speeds reads while accepting a known trade-off?

Target word: design change for read speed **Decides:** Read-heavy report on data that updates once a day.

A	Absolute language	Planned redundancy is a standard choice for reporting workloads.
B	Inverted operation	More normalization adds joins and slows reads.
C	Correct	Fewer joins speed reads; the once-a-day refresh keeps redundancy manageable.
D	Camouflage	The hardware age is not what makes seven joins slow.

4. Answer D Concept 3

A clinic database has Patients(PatientID primary key, Name) and Visits(VisitID primary key, PatientID, VisitDate). ~~The clinic sees about 90 patients a day and uses a cloud-hosted server.~~ A clerk tries to insert a visit with PatientID 7731, which does not exist in Patients. Visits.PatientID is a foreign key. What happens?

Target word: effect of the foreign key **Decides:** The referenced parent row does not exist.

A	Confusion set	The database does not silently swap in NULL for a bad value.
B	Dropped element	Foreign keys are enforced on child inserts and updates too.
C	Adjacent category	Cascade rules apply to updates and deletes of parents; they do not invent parent rows.
D	Correct	A foreign key value must match an existing parent key.

5. Answer B Concept 4

Customers has 500 rows. Orders has 1,200 rows placed by 420 distinct customers, all of whom exist in Customers. ~~The marketing team plans to email everyone after the holiday sale, which runs 10 days.~~ How many rows does Customers LEFT JOIN Orders ON CustomerID return?

Target word: left join row count **Decides:** 1,200 matched rows plus 80 customers with no orders.

A	Inverted operation	Adds the two tables as if stacking them, which is not what a join does.
B	Correct	1,200 order rows + (500 – 420) = 80 unmatched customers kept with NULLs.
C	Adjacent category	That is the inner join; it drops customers with no orders.
D	Dropped element	Assumes one row per customer, ignoring customers with several orders.

6. Answer C Concept 4 • transfer

An INNER JOIN of Products (300 rows) to Sales on ProductID returns 2,000 rows. A LEFT JOIN from Products to Sales returns 2,045 rows. How many products have never sold?

Target word: solve for unmatched rows **Decides:** Left join minus inner join = unmatched left rows.

A	Confusion set	Inner joins drop products with no sales.
B	Inverted operation	300 – 45 counts the products that did sell.
C	Correct	Each unsold product adds one NULL row: 2,045 – 2,000 = 45.
D	Wrong stem number	The left join row count, not the unmatched products.

7. Answer D Concept 5

An analyst ~~at a 3,000-seat call center~~ wants departments whose average handle time exceeds 8 minutes, counting only calls from 2025. ~~The query is scheduled to run at 6 a.m.~~ Which query structure is correct?

Target word: where each filter belongs **Decides:** Year is a row filter; average is a group filter.

A	Wrong stem number	Uses 8 as a row count instead of a threshold, and drops the year filter.
B	Inverted operation	Puts a row filter after grouping; call_year is not a grouped column.
C	Confusion set	Aggregates cannot appear in WHERE.
D	Correct	Rows are filtered before grouping; the aggregate condition goes in HAVING.

8. Answer B Concept 5 • transfer

A Tickets table has 10 rows. The Resolver column is NULL in 3 of them. What do COUNT(*) and COUNT(Resolver) return?

Target word: NULL handling in COUNT **Decides:** COUNT(column) skips NULLs.

A	Inverted operation	Swaps the two behaviors.
B	Correct	COUNT(*) counts all rows; COUNT(Resolver) counts only non-NULL values.
C	Dropped element	Ignores that COUNT(column) skips NULLs.
D	Confusion set	Counts the NULLs instead of the values.

9. Answer A Concept 6

A data team at a regional credit union ~~with 14 branches~~ needs to add a new LoyaltyTier column to the existing Members table before the third-quarter campaign. ~~The database administrator prefers work done on Tuesdays.~~ Which kind of statement and command does this require?

Target word: DDL or DML, and which command **Decides:** Adding a column changes the table's structure.

A	Correct	Changing structure is DDL; ALTER TABLE adds the column.
B	Confusion set	UPDATE changes values in a column that must already exist.
C	Inverted operation	INSERT adds rows, not columns.
D	Adjacent category	Creates a separate table instead of adding a column.

10. Answer C Concept 6 · transfer

Which statement removes only the rows whose Status is 'closed' and keeps the table and its other rows?

Target word: the command for selective row removal **Decides:** Removing some rows is DML with a condition.

A	Dropped element	Removes all rows; it takes no condition.
B	Confusion set	Removes a column, not rows.
C	Correct	DELETE with WHERE removes only matching rows.
D	Adjacent category	Removes the whole table and its structure.

11. Answer B Concept 7

A payroll system ~~at a 2,500-employee manufacturer~~ runs one transaction that debits the company account \$80,000 and credits employee accounts \$80,000. ~~The system was upgraded in March.~~ Power fails after the debit but before the transaction commits. After restart, what does atomicity guarantee?

Target word: state after a failed, uncommitted transaction **Decides:** The transaction never committed.

A	Camouflage	The upgrade date does not decide transaction behavior.
B	Correct	Atomicity is all or nothing; uncommitted work is undone.
C	Confusion set	Durability protects committed work; this never committed.
D	Inverted operation	Keeps half a transaction, the exact outcome atomicity prevents.

12. Answer A Concept 8

A media startup ~~with 35 employees in two time zones~~ stores user profiles whose fields vary widely: some users have 3 attributes, others 60. The app almost always reads a whole profile by user ID. ~~The CTO previously worked at a bank.~~ Which storage approach fits best?

Target word: storage model for this data shape **Decides:** Variable structure read whole by a single key.

A	Correct	Flexible schema and whole-record reads by key match document stores.
B	Confusion set	Graphs suit relationship traversal, which is not the access pattern here.
C	Camouflage	The CTO's background does not decide fit.
D	Adjacent category	Produces a wide, mostly NULL table and schema changes for every new attribute.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Normal forms	1NF = atomic values, no repeating groups; 2NF = no partial dependency on part of a composite key; 3NF = no transitive dependency (non-key depending on another non-key).	Mixing up which form a problem violates, or thinking 2NF applies without a composite key.	A column holding lists, or a non-key column that depends on another non-key column.
2	Denormalization trade-off	Denormalizing adds planned redundancy to speed reads, which suits read-heavy, rarely updated reporting data.	Believing more normalization always improves performance, or that redundancy is always wrong.	Slow reports built on many joins over data that changes rarely.
3	Keys and referential integrity	A primary key uniquely identifies a row; a foreign key must match an existing parent key, so orphan inserts are rejected.	Thinking a foreign key auto-creates the missing parent or is checked only on delete.	An insert or update that references a parent row that does not exist.
4	Join types and row counts	INNER JOIN keeps only matches; LEFT JOIN keeps every left row, with one row per match plus one NULL-filled row for each unmatched left row.	Expecting one row per customer from a one-to-many join, or confusing inner and left results.	Table sizes plus how many rows have matches.
5	Aggregation: WHERE, HAVING, COUNT	WHERE filters rows before grouping; HAVING filters groups after aggregation. COUNT(*) counts rows; COUNT(column) skips NULLs.	Putting an aggregate in WHERE, or thinking COUNT(column) counts NULLs.	A condition on an average or total per group, or a column with NULLs.
6	DDL vs. DML	DDL (CREATE, ALTER, DROP, TRUNCATE) defines structure; DML (SELECT, INSERT, UPDATE, DELETE) works with the rows.	Using UPDATE or INSERT to change a table's structure, or DROP to remove selected rows.	Whether the change is to the table's shape or to its data.
7	Transactions and ACID	Atomicity means all statements in a transaction commit together or none do; durability protects only committed work.	Thinking a half-finished transaction is kept or finished automatically after a failure.	A failure between two related writes.
8	Relational vs. non-relational fit	Choose the storage model by data shape and access pattern: document stores for variable-structure records read by key, graph for relationships, relational for structured data with joins and strong integrity.	Assuming relational is always right, or choosing by who prefers what.	Records whose fields vary widely, or heavy relationship traversal.

Course scope (public description): Data Management builds proficiency in using both relational and non-relational databases. Topics include selection of a data storage architecture, data types, data structures, normalization and denormalization, and querying databases. Structured Query Language (SQL) topics such as Data Definition Language (DDL) and Data Manipulation Language (DML) are covered, including joins, aggregations, and

transactions. Non-relational approaches to organizing and querying data are contrasted with relational approaches to build competency in adapting data storage architectures to business needs.